

PMKSY-OI (Pradhan Mantri Krishi Sinchai Yojana - Other Interventions) Master Project Documentation

1. Executive Summary

Comprehensive documentation covering project vision, architecture, modules, database, APIs, workflows, security, deployment, frontend, backend, future roadmap and governance.

2. Project Objectives

Provide a complete digital workflow for farmer registration, orchard infrastructure assistance applications, verification, approval, subsidy processing and reporting.

3. Technology Stack

Frontend: Angular 22, Bootstrap. Backend: Java 21, Spring Boot 3.x. Database: PostgreSQL. API Documentation: Swagger/OpenAPI.

4. System Architecture

Angular UI -> REST APIs -> Spring Boot Service Layer -> Repository Layer -> PostgreSQL Database.

5. User Roles

Public Farmer, Data Entry Operator, Assistant Director, Deputy Director, Administrator.

6. Functional Modules

Home Portal, Farmer Search, Application Registration, Orchard Management, Infrastructure Selection, Cost Estimation, Document Upload, Workflow Approval, Application Status Tracking.

7. Database Design

Master Tables: Farmer, Gender, Social Category, District, Taluk, Hobli, Village, Orchard, Infrastructure. Transaction Tables: Application, Orchard Infrastructure, Documents.

8. Farmer Module

FID based farmer lookup, demographic details, location details, land extent, validation and retrieval APIs.

9. Application Workflow

Farmer enters FID -> System fetches details -> Select orchard -> Select infrastructure -> Upload documents -> Submit -> Verification -> Approval.

10. API Inventory

Farmer APIs, Orchard APIs, Infrastructure APIs, Application APIs, Document APIs, Master Data APIs.

11. Security

Role based access control, validation, audit fields, API protection, future JWT integration.

12. Frontend Design

Government-style portal, public access landing page, farmer search page, status page, department login.

13. Backend Design

Controller, Service, Repository, DTO, Entity, Exception, Payload and Config layers.

14. Deployment

Angular frontend deployment, Spring Boot service deployment, PostgreSQL configuration, environment management.

15. Reporting & Dashboards

Application counts, subsidy statistics, district reports, infrastructure-wise reports.

16. Future Enhancements

GIS integration, Bhoomi integration, Aadhaar eKYC, SMS alerts, mobile app, analytics dashboards.

Detailed Project Scope

The PMKSY-OI platform is intended to become a complete end-to-end scheme management system for Watershed Development Department operations. It supports farmer onboarding, infrastructure assistance management, subsidy calculation, workflow approvals, document management, and reporting.